Homework: data visualization

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Based on the analysis of team statistics from Major League Baseball (MLB) from 2003 to 2023, it was found that a team's winning rate—defined as Win (W) / (Win (W) + Loss (L))—shows a certain level of correlation with both pitching and hitting metrics. The relevant data are as follows:

In pitching players:
ERA (防禦率), SV (救援成功), SVO (救援機會), H (被安打), R (被得分), ER (責失分), BB (保送), WHIP (每局被上壘率), AVG (被打擊率), TBF (投球人次), NP (投球數), P/IP (每局投球數)

In hitting players:
HR (全壘打), BB (被保送), GIDP (雙殺打)

From these findings, it is evident that pitching statistics play a crucial role in determining a team's winning percentage! Furthermore, the analysis surprisingly reveals that the batters' batting average does not impact the winning percentage as much as expected. Therefore, we will use data visualization and charts to better understand the story behind these numbers.

1. Line plot and violin plot (2 charts):
Observe the changes in the line plot for the years 2003–2023: team winning rate sorted from lowest to highest (X-axis) vs. team batting average (Y-axis). Use the changes in the violin plot to understand the distribution relationship between team winning rate (X-axis) and Team Batting Average (Y-axis).

Note: for the team winning rate across 2003–2023, the values range from a minimum of 0.2654 to a maximum of 0.7167. Please divide the winning rate into six intervals: 0.2–0.3, 0.3–0.4, 0.4–0.5, 0.5–0.6, 0.6–0.7, and 0.7–0.8. Use these intervals to present the distribution of team batting averages for the teams falling within each range.

2. Bar plot (1 chart): (continued from question 1)
Using a bar plot, observe the relationship between team winning rate (X-axis, using the same six intervals from question 1) and the total number of home runs (Y-axis).

3. Histograms (2 charts):

Please plot two histograms to compare the differences in the distribution of earned run average (ERA) between highest winning rate and lowest winning teams:

i. First histogram: for the teams with the highest winning rate each year (extract the top-performing team's data from each year), plot the pitching ERA (X-axis) vs. the number of pitchers (Y-axis).

ii. Second histogram: for the teams with the lowest winning rate each year (extract the worst-performing team's data from each year), plot the same metrics to compare against the first histogram.

Note: For the X-axis (EraRange), please divide the ERA data into the following 11 specific intervals (bins) as the following ranges: 0.0–1.0, 1.0–2.0, 2.0–3.0, 3.0–4.0, 4.0–5.0, 5.0–6.0, 6.0–7.0, 7.0–8.0, 8.0–9.0, 9.0–10.0, and 10.0–200.0.

4. Pie Charts (2 charts): (continued from question 3)

Based on the ERA intervals (bins) identified in question 3, plot two pie charts to examine the proportion of pitchers within each interval. This will help compare the distribution of pitchers across different ERA ranges between teams with high and low winning rates.